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Voluntary hyperventilation-induced hypocapnia enhances wingate test performance without altering energy system contributions

Hasan Hüseyin Yılmaz^{1,2*}, Süleyman Ulupınar³, Fatih Kırıkcı^{1,2}, Necip Fazıl Kishalı¹, Cebirail Genççoğlu³, Serhat Özbay³ and Çağrı Çiydem^{1,2}

Abstract

High-intensity exercise challenges acid-base homeostasis, often limiting performance due to hydrogen ion (H⁺) accumulation. This study investigated the effects of voluntary hyperventilation-induced hypocapnia (a state of reduced carbon dioxide levels in the blood) on Wingate test performance and energy system contributions in team sport athletes. Seventeen male athletes performed 30-second Wingate tests under control and hyperventilation conditions, separated by 2–3 days. Hyperventilation involved 5 min of controlled breathing with a tidal volume of 1000 mL and a respiratory frequency of 30 breaths·min⁻¹ to induce hypocapnia before the test. Performance metrics (peak and mean power), energy system contributions (oxidative, glycolytic, ATP-PCr), and blood lactate accumulation ($\Delta[\text{BLa}^-]$) were assessed using breath-by-breath gas analysis and capillary blood samples. Paired t-tests revealed that hyperventilation significantly increased relative peak power (12.7 ± 1.55 vs. 11.9 ± 1.05 W·kg⁻¹, $p=0.007$) and mean power (9.0 ± 0.65 vs. 8.6 ± 0.67 W·kg⁻¹, $p=0.005$) compared to control. However, no differences were observed in energy system contributions (oxidative: $20.1 \pm 3.9\%$ vs. $20.4 \pm 2.3\%$; glycolytic: $41.1 \pm 4.7\%$ vs. $41.4 \pm 3.7\%$; ATP-PCr: $38.9 \pm 5.5\%$ vs. $38.4 \pm 4.2\%$), total energy expenditure, or $\Delta[\text{BLa}^-]$ (12.6 ± 1.9 vs. 12.8 ± 1.2 mmol·L⁻¹). These findings suggest that hyperventilation enhances anaerobic performance without altering energy metabolism, potentially through improved buffering capacity, neuromuscular activation, or oxygen redistribution. All values are reported as mean $\pm$ standard deviation (SD). This endogenous strategy offers a practical approach to optimize short-duration, high-intensity efforts, though further research is needed to refine its application across varied protocols and populations.

Keywords Anaerobic performance, Respiratory alkalosis, Acid-base regulation, Energy system partitioning, Wingate test, Neuromuscular activation

*Correspondence:

Hasan Hüseyin Yılmaz
hasanh.yilmaz@atauni.edu.tr

¹Faculty of Sports Sciences, Atatürk University, Erzurum, Turkey

²Sport Sciences Application and Research Center, Atatürk University, Erzurum, Turkey

³Faculty of Sports Sciences, Erzurum Technical University, Erzurum, Turkey



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Introduction

High-intensity exercise poses a significant challenge to the maintenance of acid-base homeostasis in human physiology [1, 2]. During maximal or supramaximal efforts, particularly when anaerobic energy pathways such as glycolysis predominate, the rapid accumulation of hydrogen ions (H^+) in skeletal muscle leads to a decline in intracellular pH, resulting in metabolic acidosis [3–6]. This acidic state disrupts muscle contraction mechanisms by impairing calcium release from the sarcoplasmic reticulum and its binding to troponin and may interfere with glycolytic throughput by destabilizing key enzymes involved in ATP production [7]. Although physiological buffering systems—including bicarbonate (HCO_3^-), hemoglobin, and phosphate buffers—along with respiratory and renal regulatory mechanisms strive to neutralize this acidic load, they are often insufficient during short-duration, intense exercise [8]. Consequently, the acidic muscle milieu exacerbates fatigue perception, reduces mechanical performance, and compromises the sustainability of repeated efforts, posing a critical limitation in activities such as team sports that rely on high-intensity, intermittent exertion [3, 4, 9].

Voluntary hyperventilation has long been recognized as a method to modulate acid-base balance, with early studies by Nielsen & Smith [9] and Margaria et al. [4] highlighting its physiological implications. To counteract the deleterious effects of an acidic muscle environment, various interventions have been explored to enhance buffering capacity and maintain performance during high-intensity exercise. Sodium bicarbonate supplementation has been widely studied as an exogenous method to buffer acidosis [10–12]. Alternatively, an endogenous approach, voluntary hyperventilation, offers a practical and equipment-free method to induce respiratory alkalosis by reducing arterial carbon dioxide pressure ($PaCO_2$), thereby elevating blood pH [13–15]. Hyperventilation-induced hypocapnia promotes a favorable transmembrane gradient for H^+ removal, potentially mimicking or enhancing the buffering effects of bicarbonate ingestion without associated gastrointestinal discomfort [16, 17]. This hypocapnic state may facilitate transmembrane H^+ efflux by enhancing the gradient between intracellular and extracellular compartments, potentially contributing to a more favorable acid-base environment during exercise [17, 18]. Unlike exogenous interventions, voluntary hyperventilation provides athletes with a flexible, cost-effective strategy that can be initiated and terminated at will, making it particularly appealing for practical application in sports settings where rapid adjustments to physiological demands are required.

Prior studies have explored the effects of voluntary hyperventilation-induced hypocapnia on performance and metabolic responses during high-intensity exercise,

yielding heterogeneous findings. Fujii, Tsuchiya [19] demonstrated that pre-exercise hyperventilation reduced oxygen uptake during a 30-second Wingate anaerobic test without impairing mechanical performance, suggesting a compensatory increase in anaerobic metabolism. Similarly, Dobashi et al. [16, 17] showed that both short (5 min) and prolonged (20 min) hyperventilation attenuated aerobic energy contribution and elevated anaerobic energy reliance during Wingate efforts, yet without affecting peak or mean power output. In contrast, Sakamoto et al. [15, 20, 21] reported improvements in repeated sprint or resistance exercise performance, attributing the benefits to enhanced acid-base buffering, preserved phosphocreatine resynthesis, and attenuation of central fatigue through mechanisms such as the Setchenov phenomenon [15, 20, 21]. These discrepancies may be attributed to differences in exercise modality, intervention duration, and recovery conditions. Thus, while evidence supports that hyperventilation-induced hypocapnia modulates metabolic pathways during high-intensity exercise, its precise impact on mechanical performance in isolated, single-bout maximal anaerobic efforts like the Wingate test remains incompletely understood.

In short-duration, high-intensity exercise such as the Wingate anaerobic test, performance is predominantly underpinned by the contributions of anaerobic energy systems—namely, the adenosine triphosphate-phosphocreatine (ATP-PCr) system and glycolysis—supplemented by a minor oxidative component. The ATP-PCr system provides immediate energy through rapid hydrolysis, while glycolysis sustains effort by generating ATP via pyruvate conversion to lactate, albeit at the cost of H^+ accumulation that can impair muscle function [22, 23]. Although the oxidative system contributes minimally during a single 30-second effort, its role becomes more prominent during recovery or repeated bouts where energy restoration is critical [24, 25].

Voluntary hyperventilation-induced alkalosis may help maintain glycolytic throughput and muscle contractility by stabilizing intracellular pH during intense exertion [14, 20]. Nonetheless, as highlighted by previous research, excessive respiratory alkalosis could impair oxygen delivery via cerebral and peripheral vasoconstriction, emphasizing the need for a balanced intervention tailored to exercise type and intensity [15]. Taken together, although previous studies have examined the effects of voluntary hyperventilation on performance and metabolic responses during short-duration efforts, only a few have simultaneously quantified ATP-PCr, glycolytic, and oxidative contributions under both control and hypocapnic conditions [17, 26]. Notably, those studies typically involved smaller sample sizes, non-athletic participants, or longer and less practical hyperventilation protocols.

In contrast, the present study employed a sport-specific sample of trained athletes, a brief (5-minute) hyperventilation intervention, and a crossover design to more accurately simulate and evaluate the physiological demands of competitive anaerobic efforts.

Given the critical interplay between acid-base homeostasis, energy metabolism, and performance during high-intensity exercise, this study aims to investigate the effects of voluntary hyperventilation-induced hypocapnia on Wingate test performance and energy system contributions in team sport athletes. While the potential of hyperventilation to alleviate the acidic muscle milieu has been proposed, its impact on mechanical output and the underlying metabolic pathways in a single, maximal anaerobic effort remains largely unexplored. To the best of our knowledge, while some prior studies have estimated energy system contributions following hyperventilation, none have simultaneously quantified the proportional involvement of the ATP-PCr, glycolytic, and oxidative systems under both control and hypocapnic conditions. The present study employed a rigorously standardized crossover design with trained athletes and a validated modeling approach specifically tailored to high-intensity anaerobic performance. This investigation seeks to elucidate whether such an endogenous intervention can serve as a practical strategy to optimize short-duration, high-intensity performance, offering insights into its physiological mechanisms and applicability in sports requiring repeated anaerobic efforts.

Methods

Participants

Seventeen male team sport athletes (age: 22.6 ± 1.9 years; height: 179.5 ± 6.7 cm; body mass: 73.2 ± 6.4 kg) volunteered to participate in this study. All participants were actively engaged in competitive team sports (e.g., soccer, basketball, or handball) and had maintained a consistent training regimen of 6–9 h per week for at least the preceding three years, ensuring a homogenous cohort with well-developed anaerobic capacities. Participants were thoroughly briefed on the study's purpose, procedures, potential risks (e.g., transient dizziness or discomfort from hyperventilation), and benefits (e.g., contribution to sports science), after which they provided written informed consent. The study protocol adhered to the ethical principles outlined in the Declaration of Helsinki and was approved by the Ethics Committee of Atatürk University (Approval No: 126; Date: 18.11.2022). To minimize confounding variables, participants were instructed to maintain their habitual dietary intake and refrain from strenuous physical activity for 48 h prior to each testing session throughout the study period. Exclusion criteria included: [1] any medical contraindications to maximal effort testing (e.g., cardiovascular or respiratory

conditions), [2] current or recent orthopedic injuries that could impair cycling performance, [3] use of nutritional supplements or ergogenic aids (e.g., creatine, caffeine, or bicarbonate) within the past four weeks, and [4] ongoing medical treatment that might influence metabolic or performance outcomes.

A post-hoc power analysis was conducted using GPower 3.1*, a statistical software commonly used for calculating the required sample size and achieved power in hypothesis testing [27], employing a two-tailed test with $\alpha = 0.05$, based on previously reported effect sizes of hyperventilation-induced performance improvements [27]. Sakamoto et al. [21] reported large effect sizes of Cohen's $d = 0.84$ for bench press and $d = 1.14$ for leg press performance following pre-exercise hyperventilation. Additionally, Leithäuser et al. [26], who employed a protocol more consistent with the present study's anaerobic exercise modality and measurement approach, reported an effect size of $d = 0.78$ for mean power during a 30-second Wingate test [26]. Based on these effect sizes, the achieved statistical power with 17 participants was calculated as 0.90 (for $d = 0.84$), 0.99 (for $d = 1.14$), and 0.86 (for $d = 0.78$). As all values exceeded the commonly accepted threshold of 0.80, the sample size was considered sufficient to detect meaningful differences.

Procedures

A counterbalanced crossover design was used to compare the metabolic outputs of the Wingate test (WAnT) between control and hyperventilation conditions after the current competition season. All participants performed WAnT twice once following a familiarization session allowing performing the exercises at the submaximal intensity and wearing the gas analyzer mask. Tests were conducted using a common approach of a 30-second all-out effort in both control and hyperventilation conditions. The conditions were separated by 2–3 days and completed within 7 days. Participants performed a general standardized warm-up protocol on the ergometer for 10 min consisting of pedaling at 60–70 rpm and three 3 s cycling sprints. High-intensity actions were minimized to avoid blood lactate accumulation [BLa] and any fatigue effects.

Participants performed 5 min of voluntary hyperventilation with the aim of reducing end-tidal CO_2 pressure (PETCO_2) to below 25 mmHg, consistent with previous protocols designed to induce hypocapnia, terminating the breathing protocol immediately before the 30-second Wingate test. Participants were instructed to maintain a tidal volume of 1000 mL and respiratory frequency of 30 breaths min^{-1} . Visual feedback from the computer display was used to follow respiratory volume and frequency. This respiratory pattern was preferred based on the findings of previous studies, which showed that it

produced hypocapnia but did not cause severe discomforts such as strong tetany or dizziness [28]. After the 5-min hyperventilation intervention, participants performed the WAnT and tried to maintain peak revolutions throughout the 30 s. Pedal resistance was set at 10% of body mass (previously calculated), as this higher resistive load is recommended for trained athletes to maximize anaerobic power output, whereas lower loads (typically of body mass) are used in untrained individuals due to their reduced neuromuscular efficiency and anaerobic conditioning [19, 29, 30]. Following both tests, participants breathed freely during the 30-s WAnT and 15-min post-exercise recovery.

Anthropometric measurements were conducted using standardized methods on the first test day. The height was measured with a portable stadiometer (Holtain Ltd., Crosswell, Crymch, Dyfed, United Kingdom). Body mass and body fat were assessed using air displacement plethysmography (Bod Pod, Cosmed; Chicago, IL, USA) calibrated according to the manufacturer's guidelines. A breath-by-breath oxygen consumption ($\dot{V}O_2$) measurement was carried out using a portable metabolic gas analyzer (K5b², Cosmed, Rome, Italy) 10 min pre-exercise, during the test, and 15 min post-exercise. Pre-exercise and in the 1st, 3rd, 5th, 7th, and 10th min post-exercise, blood lactate concentration was measured from capillary blood samples drawn from the fingertip of the left hand using a portable hand analyzer (Lactate Pro, Arkray, Japan). The laboratory temperature and humidity were kept constant during all test days at $\sim 21^\circ\text{C}$ and $\sim 45\%$.

The contribution of oxidative, glycolytic, and ATP-PCr energy pathways were calculated by the PCr-LA- $\dot{V}O_2$ method [25, 31–33]. Oxygen uptake ($\dot{V}O_2$) were measured during the rest (10 min), exercise, and recovery (15 min) phases using a portable gas-exchange system in breath-by-breath mode. Prior to each test, the portable metabolic gas analyzer was calibrated according to the manufacturer's guidelines. Blood lactate concentrations were measured from capillary blood samples drawn from the fingertip of the left hand using a portable hand analyzer. Blood samples (20 μL) were collected at rest and following the exercises (first, third, fifth, seventh, and 10th min of recovery) to determine peak [BLa]. The delta lactate concentration ($\Delta[\text{BLa}]$) was determined by the difference between peak and baseline values. To avoid any possible effects of perspiration on the measurements, the fingertip was cleaned with alcohol and dried immediately before each measurement. The oxidative pathway contribution was calculated from $\dot{V}O_2$ above resting levels during the tests. The resting $\dot{V}O_2$ levels was determined prior to all tests, the average of the last five minutes of the 10 min measurement was accepted.

Phosphagen energy contribution was determined from the fast component of excess post-exercise oxygen

consumption (EPOC), reflecting ATP and phosphocreatine (PCr) resynthesis during recovery [25, 31, 32, 34]. This was modeled using a bi-exponential fit to EPOC curves, implemented through a custom-developed Python-based algorithm, alongside OriginPro software (v9.0, OriginLab, USA) as a reference [33]. The Python algorithm enhanced this process by integrating user-defined baseline $\dot{V}O_2$ values, allowing for a more precise alignment of recovery kinetics with individual physiological endpoints. The fast phase was isolated by subtracting the slow-phase $\dot{V}O_2$ from the total, with the assumption that aerobic energy fully drives PCr replenishment, assuming minimal glycolytic or myoglobin rebinding effects. This approach, facilitated by the Python algorithm's adaptive curve fitting and physiologically informed parameter bounds, offered advantages such as improved accuracy in estimating alactic energy contributions and reduced errors from auto-calibrated baselines. EPOC kinetics were fitted using the bi-exponential equation:

$$\dot{V}O_{2(t)} = \dot{V}O_{2\text{baseline}} + A_1 \left[e^{-(t-t_d)/\tau_1} \right] + A_2 \left[e^{-(t-t_d)/\tau_2} \right] \quad (1)$$

The oxidative, glycolytic, and ATP-PCr pathway contributions, as well as total energy expenditure (TEE), were calculated using a combination of validated protocols and modeling equations derived from previous publications [6, 31, 32, 35]. Liter oxygen values obtained from each energy pathway were presented in kilojoules by multiplying by 20.9, which represents the caloric equivalent of 1 L of oxygen under mixed substrate oxidation (approximately 5 kcal or 20.9 kJ per liter O_2), and expressed as absolute terms and relative percentages of total metabolic work.

Statistical analysis

Exponential models for EPOC kinetics were generated using OriginPro (v2019b, OriginLab, Northampton, MA, USA) as a reference tool, complemented by a custom-developed Python-based algorithm that incorporated user-defined $\dot{V}O_{2\text{baseline}}$ values for enhanced modeling accuracy. The data were analyzed using the Statistical Package for the Social Sciences version 27.0 (IBM Corp, Chicago, IL, USA). Data normality was verified using the Kolmogorov–Smirnov test. Performance variables, physiological responses, and energy pathway contributions were compared using the paired t-test to control the differences between control and hyperventilation conditions. The alpha level was set at 0.05 for all analyses. The data are reported as mean $\pm$ standard deviation.

Table 1 Performance outputs and metabolic results for control vs. hyperventilation conditions

	Control	Hyperventilation	t-value	p-value
Relative peak power ($\text{W}\cdot\text{kg}^{-1}$)	11.9 ± 1.05	12.7 ± 1.55	3.124	0.007
Relative mean power ($\text{W}\cdot\text{kg}^{-1}$)	8.6 ± 0.67	9.0 ± 0.65	3.276	0.005
Oxidative (kJ)	26.5 ± 2.3	25.8 ± 3.7	1.093	0.291
Glycolytic (kJ)	54.1 ± 6.1	53.5 ± 10.3	0.318	0.755
ATP-PCr (kJ)	51.1 ± 10.7	51.8 ± 14.7	0.180	0.860
Oxidative (%)	20.4 ± 2.3	20.1 ± 3.9	0.434	0.670
Glycolytic (%)	41.3 ± 3.7	41.0 ± 4.7	0.235	0.817
ATP-PCr (%)	38.3 ± 4.2	38.9 ± 5.5	0.304	0.765
TEE (kJ)	131.9 ± 15.7	131.1 ± 22.6	0.176	0.862
[BLa] ($\text{mmol}\cdot\text{L}^{-1}$)	13.7 ± 1.2	13.5 ± 1.9	0.538	0.598
$\Delta[\text{BLa}]$ ($\text{mmol}\cdot\text{L}^{-1}$)	12.8 ± 1.2	12.6 ± 1.9	0.454	0.656
VO_2 ($\text{L}\cdot\text{min}^{-1}$)	2.54 ± 0.22	2.47 ± 0.36	1.093	0.291

TEE, total energy expenditure; $\Delta[\text{BLa}]$, change in blood lactate concentration. Note: * $p < 0.05$

Results

The performance outputs and metabolic responses for the control and hyperventilation conditions during the Wingate test are presented in Table 1. End-tidal CO_2 pressure (PETCO_2) significantly decreased from 38.7 ± 3.6 mmHg at rest to 24.3 ± 3.2 mmHg following the hyperventilation protocol ($p < 0.001$, $n = 17$), confirming the successful induction of a hypocapnic state. Paired t-tests revealed that voluntary hyperventilation significantly enhanced anaerobic performance. Specifically, relative peak power was significantly higher in the hyperventilation condition ($12.7 \pm 1.55 \text{ W}\cdot\text{kg}^{-1}$) compared to the control condition ($11.9 \pm 1.05 \text{ W}\cdot\text{kg}^{-1}$; $t_{(16)} = 3.124$, $p = 0.007$). Similarly, relative mean power was significantly greater following hyperventilation ($9.0 \pm 0.65 \text{ W}\cdot\text{kg}^{-1}$) than in the control condition ($8.6 \pm 0.67 \text{ W}\cdot\text{kg}^{-1}$; $t_{(16)} = 3.276$, $p = 0.005$). These improvements indicate that hyperventilation-induced hypocapnia effectively enhanced both peak and sustained power outputs during the Wingate test.

In contrast, no significant differences were observed in the metabolic contributions of the energy systems between the two conditions. The oxidative contribution was 25.8 ± 3.7 kJ in the hyperventilation condition compared to 26.5 ± 2.3 kJ in the control condition ($t_{(16)} = 1.993$, $p = 0.291$). The glycolytic contribution was 53.5 ± 10.3 kJ in the hyperventilation condition and 54.1 ± 6.1 kJ in the control condition ($t_{(16)} = 0.318$, $p = 0.755$). Similarly, the ATP-PCr contribution was 51.8 ± 14.7 kJ in the hyperventilation condition compared to 51.1 ± 10.7 kJ in the control condition ($t_{(16)} = 0.180$, $p = 0.860$). When expressed as percentages, the relative contributions of the energy systems also showed no significant differences: oxidative ($20.1 \pm 3.9\%$ vs. $20.4 \pm 2.3\%$; $t_{(16)} = 0.434$, $p = 0.670$), glycolytic ($41.1 \pm 4.7\%$ vs. $41.4 \pm 3.7\%$; $t_{(16)} = 0.235$, $p = 0.817$), and ATP-PCr ($38.9 \pm 5.5\%$ vs. $38.4 \pm 4.2\%$; $t_{(16)} = 0.304$, $p = 0.765$) for hyperventilation and control conditions, respectively. Total energy expenditure (TEE) was also similar between conditions, with 131.1 ± 22.6 kJ in the

hyperventilation condition and 131.9 ± 15.7 kJ in the control condition ($t_{(16)} = 0.176$, $p = 0.862$).

Additionally, neither the change in blood lactate concentration ($\Delta[\text{BLa}]$) nor the absolute post-exercise lactate values differed significantly between conditions. $\Delta[\text{BLa}]$ was $12.6 \pm 1.9 \text{ mmol}\cdot\text{L}^{-1}$ in the hyperventilation condition and $12.8 \pm 1.2 \text{ mmol}\cdot\text{L}^{-1}$ in the control condition ($t_{(16)} = 0.454$, $p = 0.656$), while absolute peak [BLa] values were $13.5 \pm 1.9 \text{ mmol}\cdot\text{L}^{-1}$ and $13.7 \pm 1.2 \text{ mmol}\cdot\text{L}^{-1}$, respectively ($t_{(16)} = 0.538$, $p = 0.598$). Furthermore, absolute oxygen consumption (VO_2) during the Wingate test did not significantly differ between trials (2.47 ± 0.36 vs. $2.54 \pm 0.22 \text{ L}\cdot\text{min}^{-1}$; $t_{(16)} = 1.093$, $p = 0.291$).

Discussion

The present study demonstrated that voluntary hyperventilation-induced hypocapnia significantly enhanced Wingate test performance, as evidenced by increases in relative peak power and relative mean power. However, no significant changes were observed in the relative contributions of the oxidative, glycolytic, or ATP-PCr energy systems, nor in total energy expenditure (TEE) or blood lactate accumulation ($\Delta[\text{BLa}]$). These findings support our hypothesis that hyperventilation enhances anaerobic performance without altering the underlying energy system contributions. This dissociation between improved output and stable metabolic contribution suggests that the ergogenic effects of hyperventilation may stem from mechanisms other than changes in energy metabolism, such as altered acid-base balance [10, 36]. Specifically, voluntary hyperventilation-induced alkalosis may have optimized intramuscular conditions by delaying the onset of fatigue-related acidosis, thus preserving excitation-contraction coupling and enhancing force production efficiency without increasing total energy cost [20]. Additionally, hyperventilation may have reduced the work of breathing during high-intensity efforts, redirecting oxygen and energy from respiratory muscles to locomotor

muscles, thereby supporting higher output without additional metabolic demand [19]. Another plausible contributor is the enhancement of central nervous system excitability and motor unit recruitment through cerebral alkalosis [37]. Collectively, these mechanisms provide a plausible explanation for how hyperventilation enhanced performance independently of shifts in oxidative, glycolytic, or ATP-PCr pathways [25, 34]. These results contribute to the growing body of literature on respiratory strategies for enhancing high-intensity exercise performance and highlight directions for future mechanistic investigations.

Our findings are further contextualized by those of Leithäuser et al. [26], who also examined the effects of hyperventilation during a 30-second Wingate test. Similar to our results, they found no changes in the contributions of the oxidative or ATP-PCr systems under hyperventilation. However, they reported a ~9% increase (approximately 7 kJ) in glycolytic energy contribution, which was not observed in our study. This divergence may stem from differences in participant characteristics. The present sample consisted of trained male team-sport athletes with potentially superior buffering capacity and neuromuscular coordination, which may have mitigated detectable shifts in glycolytic flux. Thus, while both studies support the notion that hyperventilation can enhance mechanical output, our findings suggest that such improvements may occur independently of increased glycolytic energy release.

Building on these comparative findings, voluntary hyperventilation induces hypocapnia by reducing arterial partial pressure of carbon dioxide (PaCO_2), thereby shifting the acid-base balance toward alkalosis (elevated blood pH) [14, 17, 18]. This physiological alteration likely contributed to the observed improvements in Wingate test performance. The increases in relative peak power (6.7%) and mean power (4.7%) following hyperventilation may be explained by an improved acid-base status due to reduced carbonic acid concentration, which facilitates hydrogen ion efflux and delays acidosis during high-intensity effort. Since the Wingate test relies heavily on anaerobic glycolysis, which generates H^+ as a byproduct, respiratory alkalosis may delay the onset of intramuscular acidosis, thereby preserving muscle contractility and force output [25, 34]. In addition, hypocapnia induces a leftward shift in the oxyhemoglobin dissociation curve, increasing hemoglobin's affinity for oxygen. Although this shift may slightly elevate arterial oxygen saturation during the initial seconds of the test, arterial saturation is already near-maximal in healthy individuals at rest and during most exercise intensities. More critically, this shift may impair oxygen unloading at the muscular level, potentially restricting mitochondrial oxygen availability at the onset of intense muscular demand [38]. Therefore,

the net impact of hypocapnia on oxygen transport is context-dependent—balancing improved pulmonary loading with possible reductions in peripheral extraction. Moreover, hyperventilation may enhance central nervous system function by increasing cerebral alkalosis, which could facilitate motor unit recruitment and neural drive [37, 39]. Altogether, these factors suggest that hyperventilation-induced hypocapnia may provide an ergogenic benefit by optimizing both intramuscular conditions and neuromuscular activation during short, maximal efforts.

The increase in peak power output following hyperventilation likely reflects mechanisms distinct from those contributing to improvements in average power. Peak power, generated within the first few seconds of the Wingate test, is primarily supported by the ATP-PCr system and is highly dependent on neuromuscular activation [34, 40]. The cerebral alkalosis induced by hyperventilation may enhance central nervous system excitability and facilitate motor unit recruitment, promoting more rapid and forceful muscle activation [37, 39]. Enhanced synchronization and activation of high-threshold motor units could account for the observed peak power gains, even in the absence of detectable shifts in energy system engagement [37]. Additionally, improved intramuscular buffering might delay inhibitory effects of early H^+ accumulation—though acidosis likely plays a minimal role during such a brief initial effort phase [20, 25]. In contrast, the improvement in mean power output over the 30-second Wingate test likely reflects increased fatigue resistance and more sustained contractile function under conditions of respiratory alkalosis. As anaerobic glycolysis becomes the dominant energy source throughout the test, accumulating H^+ can progressively impair force generation [7, 22]. Hyperventilation may attenuate this effect by enhancing muscle buffering capacity and promoting more efficient H^+ efflux, thereby sustaining muscle performance throughout the duration of effort [17, 37]. These physiological advantages help explain the observed enhancement in average power output, despite unchanged total energy expenditure and stable contributions from the oxidative, glycolytic, and ATP-PCr systems [25, 34].

Despite the observed improvements in performance, voluntary hyperventilation did not significantly alter the relative contributions of the oxidative, glycolytic, or ATP-PCr energy systems during the Wingate test. While this finding may initially appear counterintuitive, it can be explained by improved energy utilization efficiency rather than increased metabolic flux. Specifically, respiratory alkalosis may enhance excitation–contraction coupling and motor unit recruitment, enabling greater mechanical output per unit of ATP consumed [37, 40]. Furthermore, hyperventilation performed prior to exercise may reduce ventilatory drive during exertion, decreasing the

oxygen cost of breathing and reallocating oxygen toward locomotor muscles without affecting whole-body VO_2 [19, 41]. The 30-second duration of the Wingate test also inherently limits oxidative ATP production and may obscure subtle shifts in system-specific energy contributions. It is also plausible that small or short-lived changes in phosphagen or glycolytic flux remained below the detection threshold of post-exercise estimations. Collectively, these considerations help reconcile how performance improved in the absence of significant changes in measured metabolic contributions. Notably, the absence of change in total energy expenditure despite enhanced performance does not represent a physiological contradiction. Instead, it may reflect greater force production efficiency under conditions of respiratory alkalosis [20, 25]. Mitochondrial ATP synthesis may also become more efficient at higher pH levels, producing more ATP per unit substrate oxidized—though this hypothesis remains speculative [17, 40]. Additionally, if hyperventilation reduced the respiratory workload during maximal effort, more oxygen could be diverted from respiratory to locomotor muscles, enhancing contractile capacity without elevating systemic VO_2 [41].

Our findings partially align with those of Dobashi et al. [17], who reported a significant reduction in oxygen uptake during the Wingate test following 5-minute hyperventilation, without changes in performance. In the current study, oxygen uptake during the 30-second WAnT also tended to be lower in the hyperventilation condition ($2.47 \pm 0.36 \text{ L} \cdot \text{min}^{-1}$) compared to the control condition ($2.54 \pm 0.22 \text{ L} \cdot \text{min}^{-1}$), though the difference did not reach statistical significance. This suggests a potential reduction in aerobic metabolism, albeit not large enough to be detected at the group level. One plausible explanation for this discrepancy is the difference in participant characteristics. Our cohort consisted of trained male team-sport athletes, while Dobashi et al. [17] included a more heterogeneous sample of moderately trained and active individuals. Given that trained athletes often exhibit enhanced oxygen kinetics and buffering capacity, their physiological adaptations may buffer small metabolic shifts or mask their detectability. Nonetheless, both studies support the idea that hyperventilation can improve mechanical performance without markedly altering whole-body aerobic energy demand, underscoring the role of neural and acid-base regulatory factors.

The ergogenic effects of hyperventilation (HV) appear to be context-dependent, varying by exercise modality, protocol duration, and athlete characteristics. For instance, Sakamoto et al. [21] demonstrated that HV-assisted recovery improved repetition volume and joint velocity in resistance exercises, likely via enhanced acid-base buffering and reduced muscle acidosis. Similarly, HV attenuated performance decline during repeated

sprint pedaling, particularly in later sets [20]. However, duration is critical—Sakamoto et al. [15] found no benefit with 15-second HV, while 45 s increased ventilatory workload and decreased aerobic metabolism. Fujii et al. [19], using a 30-second Wingate test, reported decreased oxygen uptake and increased lactate following HV, yet without performance benefits. In contrast, our 5-minute HV protocol improved power output without significantly altering VO_2 or lactate. These differences likely reflect the interaction between HV duration, exercise type, and participant training status, with longer HV (as in our study) perhaps maximizing alkalotic benefits without provoking adverse ventilatory effects. Such evidence emphasizes the importance of tailoring HV protocols to the specific metabolic and neuromuscular demands of a given task.

The lack of significant changes in energy system contributions despite enhanced performance may be attributed to the intrinsic characteristics of the Wingate test. The 30-second duration relies primarily on anaerobic pathways, with minimal oxidative contribution. Both the ATP-PCr ($\sim 38\%$) and glycolytic ($\sim 41\%$) systems support rapid ATP turnover and are less responsive to transient alkalosis within the physiological range. For example, PFK-1, a rate-limiting enzyme in glycolysis, may require more profound pH shifts to be inhibited or enhanced [7]. Moreover, our observed $\Delta[\text{BLa}^-]$ values (12.6 ± 1.9 vs. $12.8 \pm 1.2 \text{ mmol} \cdot \text{L}^{-1}$) suggest that systemic lactate dynamics remained stable across trials. The stable blood lactate levels observed across conditions suggest that while respiratory alkalosis may influence intramuscular H^+ dynamics, it does not necessarily translate into reduced systemic lactate accumulation. Methodological limitations should also be acknowledged. Our use of only male team-sport athletes may limit generalizability, and the reliance on post-exercise estimations to infer energy contributions may not capture brief or subtle fluxes. Future studies should expand on this work by exploring different HV durations, including female participants, and integrating real-time intramuscular pH monitoring to better elucidate the nuanced interplay between respiratory manipulations and metabolic regulation during high-intensity efforts.

Conclusions

This study demonstrated that voluntary hyperventilation-induced hypocapnia significantly enhances Wingate test performance in team sport athletes, as evidenced by increases in relative peak power and mean power. However, no alterations were observed in the relative contributions of the oxidative, glycolytic, or ATP-PCr energy systems, nor in total energy expenditure or blood lactate accumulation ($\Delta[\text{BLa}^-]$). The ergogenic benefits observed suggest that voluntary hyperventilation could serve as a

practical, endogenous strategy for optimizing short-duration, high-intensity efforts in sports settings, particularly where rapid performance adjustments are critical. However, the lack of systemic lactate reduction and the use of a fixed 5-minute protocol highlight the need for further research to optimize hyperventilation duration and assess its effects across diverse populations and exercise modalities. Collectively, these results advance our understanding of respiratory interventions in exercise physiology and underscore their potential as a cost-effective tool for enhancing anaerobic capacity.

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

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Author contributions

HHY: Conceptualization, Data curation, Formal analysis, Funding acquisition, Writing – original draft, Writing – review & editing. SU: Conceptualization, Data curation, Formal analysis, Funding acquisition, Writing – original draft. SÖ: Conceptualization, Data curation, Formal analysis, Investigation, Resources, Writing – original draft. FK: Conceptualization, Data curation, Formal analysis, Funding acquisition, Resources, Writing – review & editing. NFK: Project administration, Conceptualization, Formal analysis, Investigation, Resources, Writing – review & editing. CG: Investigation, Methodology, Writing – review & editing. ÇÇ: Conceptualization, Data curation, Investigation, Software, Writing – review & editing.

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Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

The study received approval from the Ethics Committee of Atatürk University, Sport Sciences Faculty (Approval No: 126 Date: 18.11.2022), ensuring compliance with the ethical standards stipulated by the Declaration of Helsinki. Informed consent was obtained in written form from all participants prior to their involvement in the research, affirming their voluntary participation and comprehension of the study's procedures and objectives.

Consent for publication

No individual data is being published as part of this manuscript.

Competing interests

The authors declare no competing interests.

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